M. Abdullah Siddiqui 23I-0617 CS-F

Problem 1:-

(a)

$$y - v_1 = \begin{bmatrix} 20 \\ 0 \\ 4 \end{bmatrix} \quad v_2 - v_1 = \begin{bmatrix} 3 \\ -5 \\ 1 \end{bmatrix} \quad v_3 - v_1 = \begin{bmatrix} 7 \\ -3 \\ 5 \end{bmatrix}$$

$$c_1(v_2 - v_1) + c_2(v_3 - v_1) = y - v_1$$

$$\left[\begin{array}{cc|c} 3 & 7 & 20 \\ -5 & -3 & 0 \\ 1 & 5 & 4 \end{array}\right] \sim \begin{bmatrix} 1 & 0 & -30/13 \\ 0 & 1 & 50/13 \\ 0 & 0 & -168/13 \end{bmatrix}$$

$$c_1 = \frac{-30}{13} \quad c_2 = \frac{50}{13}$$

$$y - v_1 = \frac{-30}{13} v_2 + \frac{30}{13} v_1 + \frac{50}{13} v_3 - \frac{50}{13} v_1$$

$$y = \frac{-30}{13} v_2 + \frac{50}{13} v_3 - \frac{7}{13} v_1$$

(b)

$$\tilde{v}_1 = \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix} \quad \tilde{v}_2 = \begin{bmatrix} 0 \\ -4 \\ 2 \\ 1 \end{bmatrix} \quad \tilde{v}_3 = \begin{bmatrix} 4 \\ -2 \\ 6 \\ 1 \end{bmatrix} \quad \tilde{y} = \begin{bmatrix} 17 \\ 1 \\ 5 \\ 1 \end{bmatrix}$$

$$c_1 \tilde{v}_1 + c_2 \tilde{v}_2 + c_3 \tilde{v}_3 = \tilde{y}$$

$$\left[\begin{array}{ccc|c} -3 & 0 & 4 & 17 \\ 1 & -4 & -2 & 1 \\ 1 & 2 & 6 & 5 \\ 1 & 1 & 1 & 1 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & -3 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 4 \end{array}\right]$$

Incons

$c_1 = -3$

$c_2 = -2$

$c_3 = 2$

$0 \neq 4$

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Problem 3 :-

(a)

$$v_2 - v_1 = \begin{bmatrix} 1 \\ 2 \\ -2 \\ 0 \end{bmatrix} \qquad v_3 - v_1 = \begin{bmatrix} 0 \\ 3 \\ -4 \\ -1 \end{bmatrix}$$

set $\{ v_2 - v_1 , v_3 - v_1 \}$

is linearly independent

(b)

$$\tilde{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 2 \\ 1 \\ 1 \end{bmatrix} \qquad \tilde{v}_2 = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 1 \\ 1 \end{bmatrix} \qquad \tilde{v}_3 = \begin{bmatrix} 1 \\ 2 \\ -2 \\ 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 \\ -1 & 1 & 2 \\ 2 & 0 & -2 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 3 \\ 0 & -4 & -4 \\ 0 & -1 & -1 \\ 0 & -1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

linearly independent

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Problem 4:-

$$\left[\begin{array}{ccc|c} \tilde{v}_1 & \tilde{v}_2 & \tilde{v}_3 & \tilde{p} \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 5 \\ -1 & 1 & 2 & 4 \\ 2 & 0 & -2 & -2 \\ 1 & 1 & 0 & 2 \\ 1 & 1 & 1 & 1 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$c_1 = -2$

$c_2 = 4$

$c_3 = -1$

Problem 5:-

let W be subspace spanned by $S = \{v_1, v_2, v_3\}$

$$\text{Proj}_W\, p_1 = \frac{p_1 v_1}{v_1 v_1} v_1 + \frac{p_1 v_2}{v_2 v_2} v_2 + \frac{p_1 v_3}{v_3 v_3} v_3$$

$$= \frac{7}{7} v_1 + \frac{7}{7} v_2 + \frac{7}{7} v_3$$

$$= \begin{bmatrix} 1 \\ -1 \\ 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \\ 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \\ 2 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3 \\ -3 \\ 6 \\ 3 \end{bmatrix} \Rightarrow 3 \begin{bmatrix} 1 \\ -1 \\ 2 \\ 1 \end{bmatrix}$$

coefficients doesn't sum up to 1 so not aff comb. & convex comb but in linear combin

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